

Receipts Pack — for Kau’s deck slides 2–3

AT3, per docs/WEEKEND_PLAN.md: “screenshots/quotes with page refs — Sarvam blog (Svarah eval, underrepresentation rationale, five aggregate metrics, no subgroup table), ASR-FAIRBENCH’s mixed-effects paragraph, the clinical audit’s system list.” All three below, with exact quotes, verified against the live source on 2026-08-29 (not from memory), plus one finding that’s sharper than what the brief asked for.

1. Sarvam Saaras V3 blog — underrepresentation rationale, Svarah “validation,” no subgroup table

Source: sarvam.ai/blogs/asr, “Introducing Saaras V3,” published February 10, 2026. Section: “**English Benchmark: Svarah.**”

“Indian-accented English is underrepresented in widely used ASR benchmarks such as LibriSpeech and Switchboard, despite India having an estimated 130 million English speakers. As a result, performance on these standard datasets does not fully reflect how systems handle English as spoken in India.

To address this gap, Saaras V3 was evaluated on Svarah, a 9.6-hour benchmark comprising 117 speakers across 65 districts in 19 Indian states. The dataset captures substantial accent variation and includes both read and spontaneous conversational speech across real-world domains, providing a more representative assessment of Indian English ASR performance.

This performance reflects the model’s strong ability to handle accent variation and phonetic shifts more reliably than systems primarily optimized on Western English benchmarks.”

Three things to point at on the slide:

1. **The underrepresentation argument is verbatim ours.** “Indian-accented English is underrepresented in widely used ASR benchmarks... despite India having an estimated 130 million English speakers” is exactly Chapter 1’s framing — shipped by a funded lab, on our exact benchmark, before we could publish it.

2. **No WER number is given for Svarah at all.** Read that Svarah section again: it validates on Svarah, describes the dataset, and asserts the model handles “accent variation... more reliably” — but never states a number. Not an aggregate WER, let alone a subgroup one. This is a *stronger* gap than “aggregate-only” — it’s “claimed, unquantified.”
3. **Where they do quantify, it’s a single number, no subgroups.** Their IndicVoices claim two paragraphs earlier — “Saaras V3 achieves a word error rate of 19.31 percent” — is one aggregate number across 10 languages, with a second sentence noting the gap “widens” on 12 more languages, but still no per-language or per-accent breakdown, and no significance test anywhere in the post.

Companion source (aggregate-metrics framing): sarvam.ai/speech-to-text, retrieved 2026-08-29:

“Sarvam also uses a 5-metric evaluation framework (WER, LLM-WER, COMET, Intent Score, Entity Preservation) instead of relying on WER alone, because for Indian languages, two transcriptions with identical WER can have very different semantic accuracy.”

Five metrics, zero subgroups, zero significance tests — even where they explicitly reject “WER alone” as insufficient, the axis they add is *metric diversity*, not *demographic-subgroup* diversity. That’s precisely the gap our audit fills.

2. ASR-FAIRBENCH — the mixed-effects Poisson paragraph

Source: A. K. Rai, S. Rahangdale, U. Anand, A. Mukherjee, “ASR-FAIRBENCH: Measuring and Benchmarking Equity Across Speech Recognition Systems,” *Interspeech 2025*, arXiv:2505.11572. Section 2.2.1, “Word error rate and fairness model” (verified against the arXiv HTML rendering, 2026-08-29):

“To assess fairness across demographic groups, we employ mixed-effects Poisson regression:

$$\log(\text{WER}_i) = \beta_0 + \beta_1 X_i + \beta_2 Z_i + u_i \quad (1)$$

where X_i represents demographic attributes, Z_i represents covariates, and β_1 quantifies disparity through disparity ratio = e^{β_1} .”

And from §2.2.3, “Statistical adjustment and FAAS”:

“Statistical significance of fairness disparities is assessed via the Likelihood Ratio Test comparing full and reduced models... The resulting p-value indicates whether disparities are statistically significant. When disparities are significant ($p < 0.05$), we apply a proportional penalty to the category score.”

Why this is the load-bearing citation for our pivot, not just a nice-to-have:

- This is *functionally the same test* we built independently: a Poisson model of WER against a demographic covariate, with a likelihood-ratio / deviance test and a $p < 0.05$ threshold for “real” disparity. We did not copy this — our Phase 2 pipeline predates knowledge of this paper — but it means we can no longer claim the statistical machinery as a contribution. We can, and should, cite it, adopt its framing explicitly, and spend our novelty budget elsewhere (the multi-system audit itself, and the Indian-accent axis it never covers).
 - **Their dataset is Fair-Speech (US English, self-reported demographics: age, gender, ethnicity, location, native-language status) — not Indian accents, and not Svarah.** Their own conclusion (§4) singles out Whisper-tiny for “better overall fairness” despite higher WER and flags that “fairness issues can outweigh slight WER improvements” for fine-tuned Wav2Vec/HuBERT — i.e., their own headline finding is exactly our thesis (“aggregate WER hides subgroup movement”), just demonstrated on a different population. **The accent-family axis on Indian English is the one gap even the paper that already published our statistical test does not fill.** That’s the sentence for the slide.
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3. The clinical audit paper — system list and axes

Source: S. Kumar, P. Shivaprakash, A. Manoharan, A. Kurariya, D. Mukherjee, L. Shukla, A. Mukherjee, P. Chand, P. Murthy, “ASR Under the Stethoscope: Evaluating Biases in Clinical Speech Recognition across Indian Languages,” arXiv:2512.10967, submitted 30 Nov 2025 (verified against the arXiv abstract page and HTML rendering, 2026-08-29).

“In this study, we conduct the first systematic audit of ASR performance on real-world clinical interview data spanning Kannada, Hindi, and Indian English, comparing leading models including IndicWhisper, Whisper-large-v3, Sarvam, Google speech-to-text, Gemma3n, Omnilingual, Vaani, and Gemini... with a particular focus on error

patterns affecting patients vs. clinicians and gender-based or intersectional disparities.”

System list, exactly as stated in the paper (8 systems): IndicWhisper, Whisper-large-v3, Sarvam, Google Speech-to-Text, Gemma3n, Omnilingual, Vaani, Gemini.

Why this is the receipt for “third-party ASR audits are a normal, publishable thing, and our axis is still open”:

- This paper **includes Sarvam** in a multi-system audit and got through peer review (submitted to arXiv Nov 2025; a related/extended companion, “SamaVaani,” arXiv:2606.26901, went further and proposed a debiasing method on top of the same audit). A third-party comparative audit of Indian ASR systems, including a commercial one, is not a novel genre we’re inventing — it’s a genre with precedent.
 - **Its subgroup axes are speaker role (doctor vs. patient) and gender**, in a clinical-interview setting, across Kannada/Hindi/Indian English. It does **not** stratify by accent or language family, and it does not touch Svarah. **The accent-family axis, on the benchmark Sarvam itself uses to validate, is still nobody’s paper.** That is our slide-3 punchline.
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How to use this on the day

Every quote above is reproduced verbatim from the live source, with the retrieval date noted, so a panel member can pull up the same page or PDF and verify it in front of you — that’s the point of a “receipts pack” instead of a paraphrase. Suggested framing for slides 2–3:

- **Slide 2 (the encroachment):** the Sarvam quote — underrepresentation rationale is verbatim ours, Svarah is their own validation benchmark, and they report *no number* for it, let alone a subgroup one.
- **Slide 3 (the literature closes the “we invented the stats” door, and opens the real gap):** ASR-FAIRBENCH already ships our exact statistical test — cite it, don’t compete with it — on a non-Indian population; the clinical audit shows third-party Indian-ASR audits are publishable, on role/gender, not accent. **Six systems, one frozen protocol, per-family WER with a p-value, on the benchmark the flagship model itself uses** is the row nobody has published — and (see audit/README.md) it is no longer a plan, it is six completed rows.